

Dark Energy: Theory, Observations, and Challenges

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Workshop on Astrophysics and Cosmology

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Homogeneous and Isotropic Universe

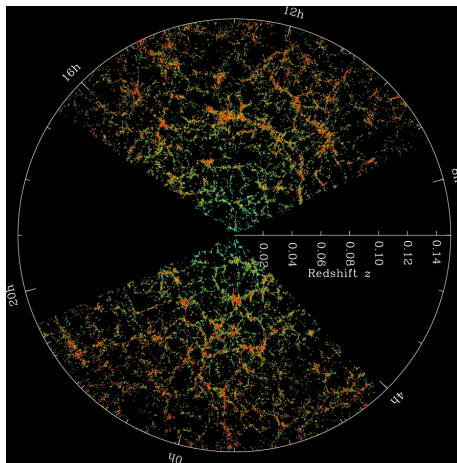


Figure: Sloan Digital Sky Survey.

Friedmann Equations

Einstein Field Equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Friedmann Equations (FLRW universe):

1. First Friedmann: $H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}$
2. Second Friedmann: $\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}$

2011 Nobel Prize in Physics

- 1999: *Riess et al.* and *Perlmutter et al.* discover that the expansion of the Universe is **ACCELERATED**

OBSERVATIONAL EVIDENCE FROM SUPERNOVAE FOR AN ACCELERATING UNIVERSE AND A COSMOLOGICAL CONSTANT

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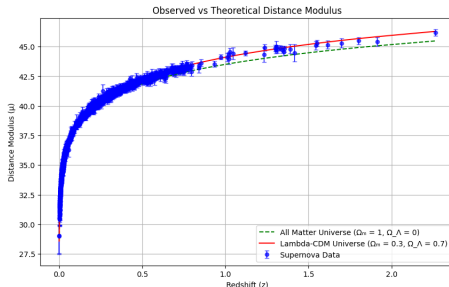
MEASUREMENTS OF Ω AND Λ FROM 42 HIGH-REDSHIFT SUPERNOVAE

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The Nobel Prize in Physics 2011

Evidence for Dark Energy

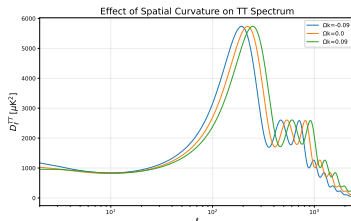
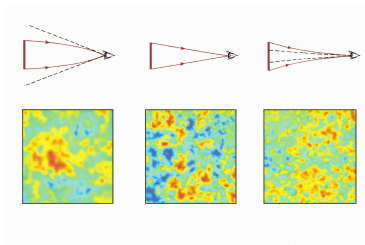


- Type Ia supernova observations show that the cosmic expansion is **accelerating**.
- Ordinary matter and gravity alone would lead to a **decelerating** expansion.
- The Λ CDM model provides the **best fit** to current supernova data.

The Λ CDM Model

- The Λ CDM model is the **standard cosmological model** describing the large-scale evolution of the Universe.
- It assumes a homogeneous and isotropic Universe governed by **General Relativity**.
- The cosmic energy budget consists of:
 - $\sim 4\%$ baryonic matter,
 - $\sim 26\%$ cold dark matter (CDM),
 - $\sim 70\%$ dark energy in the form of a cosmological constant Λ .
- Dark energy (Λ) drives the **late-time accelerated expansion** of the Universe.

CMB Angular Scale in a Flat Universe ($\Omega_k = 0$)



- Flat spatial geometry $\Rightarrow$ **straight photon geodesics**

- $\ell \simeq 220 \Rightarrow$ **Flat Universe**

$$\Omega_k = 0$$

- $\ell > 220 \Rightarrow$ **Open Universe**

$$\Omega_k > 0$$

- $\ell < 220 \Rightarrow$ **Closed Universe**

$$\Omega_k < 0$$

The angular position of the first acoustic peak provides a geometric probe of spatial curvature.

What is Λ

- Λ CDM is the **standard model of cosmology**
- Λ represents **vacuum energy (dark energy)**
- Appears in Einstein equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

- Energy density and pressure:

$$\rho_\Lambda = \frac{\Lambda}{8\pi G}, \quad p_\Lambda = -\rho_\Lambda$$

- Equation of state:

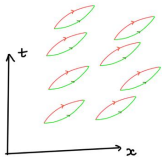
$$w_\Lambda = -1$$

- Causes **late-time accelerated expansion**
- Constant in time: $\rho_\Lambda = \text{const.}$
- Observations (SNe Ia, CMB, BAO) give:

$$\Omega_\Lambda \simeq 0.7$$

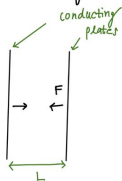
Quantum Mechanics Λ

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$



The vacuum is
not boring.

The vacuum is
a sea of virtual
particle antiparticle
particles continuously
created and
destroyed.



"Casimir Effect confirms
the existence of vacuum
fluctuation".

Boundary condition
changes & leads
to force of
Attraction.

Momentum of Particles in an Expanding Universe

Massive particles

- Momentum scales as

$$p \propto \frac{1}{a}$$

- Even if the particle is relativistic.
- Quickly,

$$p \rightarrow 0, \quad E \rightarrow mc^2$$

- Rest mass energy dominates over kinetic energy.
- On cosmic scales, massive particles effectively “hang around”.

Massless particles (photons)

- Energy–momentum relation:

$$E = pc$$

- Momentum scales as

$$p \propto \frac{1}{a}$$

- Hence,

$$E \propto \frac{1}{a}$$

- This leads to **redshift**.
- Photon Pressure : $P = \rho/3$

Equation of State

- **Equation of State (EoS):**

- Describes the physical relation between pressure and energy density

$$p = w\rho$$

- **Equation of State Parameter:**

- Defined as the ratio of pressure to energy density

$$w \equiv \frac{p}{\rho}$$

- **Cosmological Fluids:**

- **Dust (Cold Matter):**

$$p = 0 \quad \Rightarrow \quad w = 0$$

- **Radiation:**

$$p = \frac{1}{3}\rho \quad \Rightarrow \quad w = \frac{1}{3}$$

- **Vacuum Energy :**

$$p = -\rho \quad \Rightarrow \quad w = -1$$

Evolution of Energy Density

- Energy density evolution in an expanding Universe

$$\nabla_\mu T^\mu_\nu = 0 \quad \Rightarrow \quad \dot{\rho} + 3H(\rho + p) = 0$$

To be precise $\nu = 0$, gives density evolution

- Solution as a function of scale factor a :

$$\rho(a) \propto a^{-3(1+w)}$$

- **Examples:**

- **Dust (Matter):**

$$w = 0 \quad \Rightarrow \quad \rho_m \propto a^{-3}$$

- **Radiation:**

$$w = \frac{1}{3} \quad \Rightarrow \quad \rho_r \propto a^{-4}$$

- **Vacuum Energy :**

$$w = -1 \quad \Rightarrow \quad \rho_\Lambda = \text{constant} !!!$$

Second Friedmann Equation and Cosmic Acceleration

Second Friedmann (Acceleration) Equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P)$$

Pure Dust

$$P = 0$$

$$\rho > 0$$

$$\ddot{a} < 0$$

Radiation

$$P = \frac{\rho}{3}$$

$$\rho + 3P > 0$$

$$\ddot{a} < 0$$

Vacuum / Λ

$$P = -\rho$$

$$\rho + 3P = -2\rho < 0$$

$$\ddot{a} > 0$$



Negative pressure
drives acceleration

Conservation of Energy in Expanding Universe

- In classical physics, **energy is conserved**:

Time translation invariance $\implies$ Energy conservation

- In an **expanding universe**, spacetime itself evolves with time.
- Example: **Radiation energy density**

$$\rho_r \propto a^{-4} \quad \Rightarrow \quad E_r = \rho_r a^3 \propto a^{-1} \neq \text{constant}$$

- **Takeaways:**

- Energy conservation requires time-translation symmetry.
- Expanding universe breaks this symmetry.

Problems with the Cosmological Constant Λ

- **Cosmological Constant Problem**

- Quantum Field Theory predicts vacuum energy:

$$\rho_{\text{vac}}^{\text{QFT}} \sim 10^{120} \rho_{\Lambda}^{\text{obs}}$$

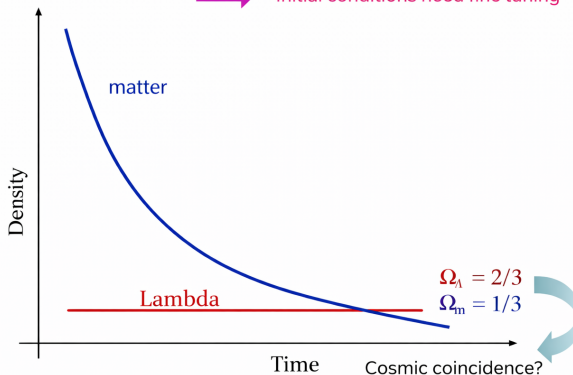
- Expected value $\Rightarrow 10^{112} \text{ ergs/cm}^3$
- Observed value $\Rightarrow 10^{-8} \text{ ergs/cm}^3$



Fine Tuning Problem.

$\rho_{\Lambda}/\rho_m \sim 10^{-44}$ at the electroweak scale if $\rho_m \sim \rho_{\Lambda}$ now.

➡ Initial conditions need fine tuning



Dynamical Dark Energy

- A scalar field respects the isotropy of the universe.
- The scalar field is taken to be a function of time only, $\phi = \phi(t)$.
- The scalar field evolves very slowly.

Have we detected any scalar field?



Higgs Field

Quintessence Model

- **Action** for a scalar field ϕ :

$$S = \int d^4x \sqrt{-g} a^3 \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right]$$

- **Energy density:**

$$\rho_\phi = \frac{1}{2} \dot{\phi}^2 + V(\phi)$$

- **Pressure:**

$$p_\phi = \frac{1}{2} \dot{\phi}^2 - V(\phi)$$

- **Equation of state:**

$$w_\phi = \frac{\frac{1}{2} \dot{\phi}^2 - V(\phi)}{\frac{1}{2} \dot{\phi}^2 + V(\phi)}$$

(can vary between $-1 < w_\phi < 1$)

Revisiting 2nd Friedmann equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho + 3p)$$

(ρ = Total energy density, p = Total pressure)

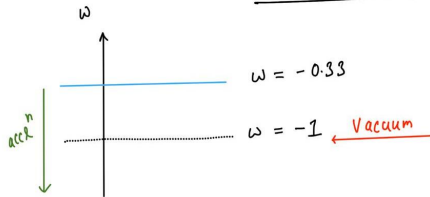
$$\ddot{a} > 0 \Rightarrow \rho + 3p < 0$$

$$\Rightarrow \rho + 3\omega\rho < 0$$

$$\Rightarrow (1 + 3\omega) < 0$$

$$\therefore \omega_{\text{eff}} < -\frac{1}{3}$$

"Accelerating universe"



Hunting for Equation of State Parameter

- **Dark energy: dynamical or constant?**

→ Number to characterize how much dark energy density evolves.

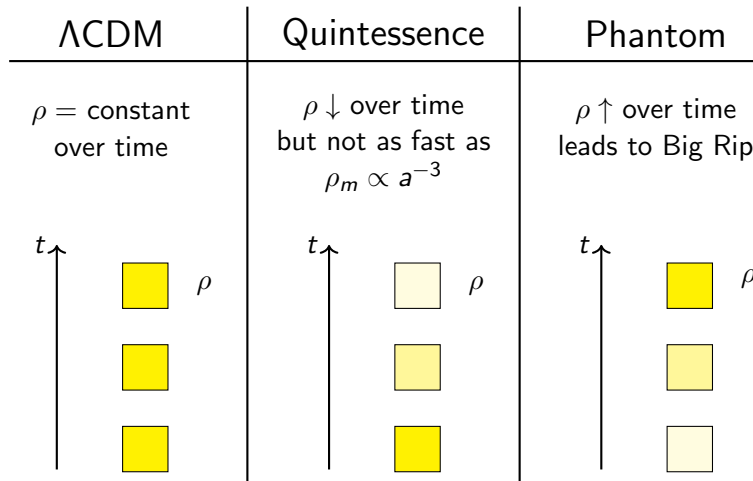
$$\rho \propto a^{-3(1+w)}$$

$$\longrightarrow w = -1 \quad (\rho = \text{constant})$$

$$\longrightarrow w > -1 \quad (\rho \propto a^{-3(1+w)}, \rho_{\text{DE}} \text{ decreasing})$$

$$\longrightarrow w < -1 \quad (\rho \propto a^{-3(1+w)}, \rho_{\text{DE}} \text{ increasing})$$

Density Evolution



Latest DESI DR2 Observation

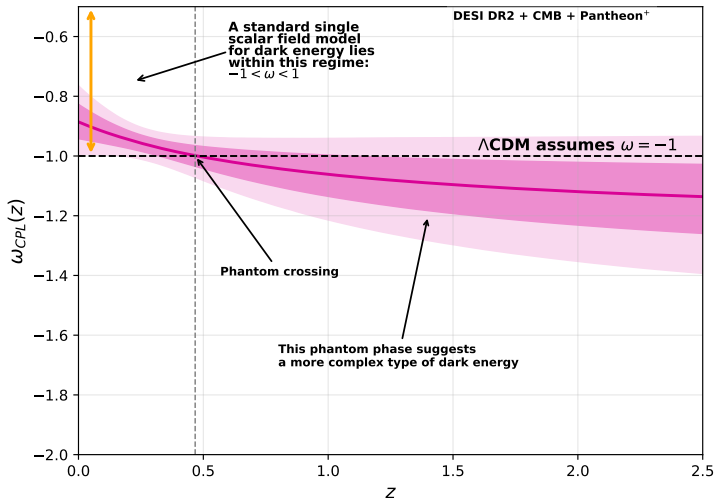
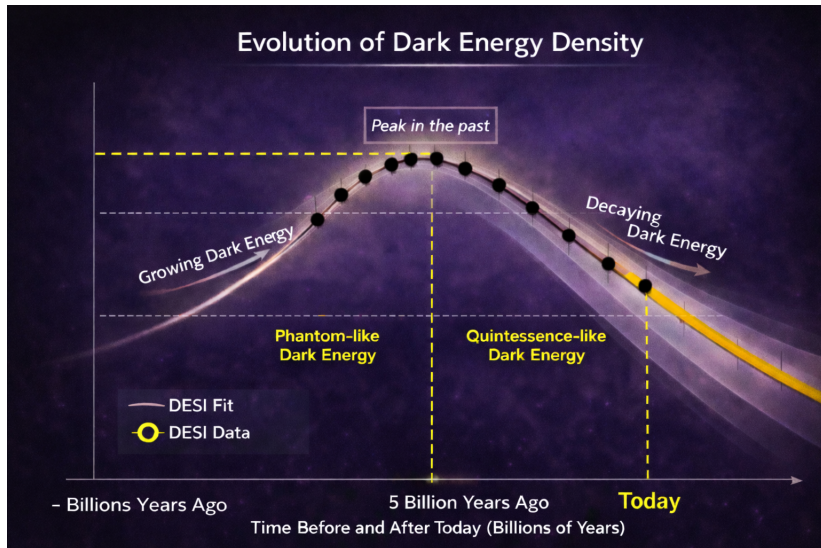
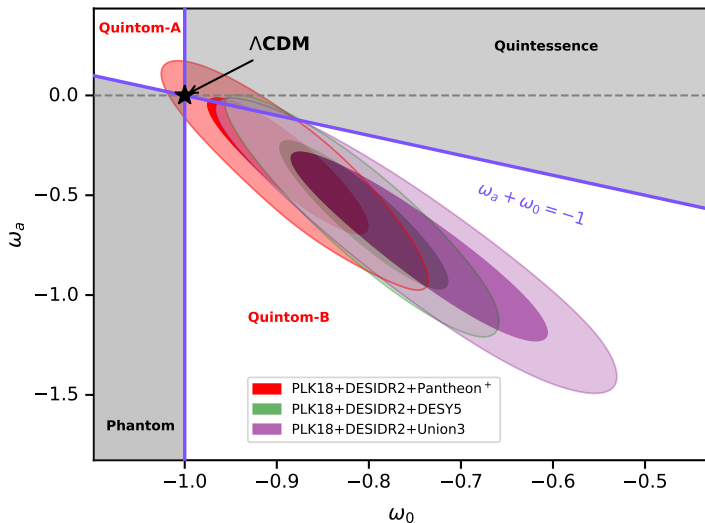


Figure: Dynamical Dark Energy

Density Evolution



w_0 - w_a Plane



Baryon Acoustic Oscillations: 13 BAO measurements from DESI DR2 in the redshift range $0.295 \leq z \leq 2.33$ *M. A. Karim et al., DESI DR2, arXiv:2503.14738 (2025).*

Cosmic Chronometers: 15 Hubble parameter measurements in the redshift range $0.1791 \leq z \leq 1.965$ *M. Moresco et al., JCAP 2012(07), 053, M. Moresco et al., MNRAS Lett. 450, L16 (2015), and M. Moresco et al., JCAP 2016(05), 014*

Pantheon⁺: 1701 light curves of 1550 SNe Ia, $0.01 \leq z \leq 2.26$. We exclude very low- z events ($z < 0.01$) where peculiar velocities dominate, leaving 1590 light curves. *D. Brout et al., Astrophys. J. 938(2), 110 (2022).*

DES-SN5Y 1829 SNe Ia, $0.025 \leq z \leq 1.12$ (1635 DES + 194 low- z anchors). *DES Collaboration et al., Astrophys. J. Lett. 973(1), L14 (2024)*

Union3: 2087 SNe Ia, $0.050 \leq z \leq 2.26$, homogenized across 24 surveys. *D. Rubin et al., Astrophys. J. 986(2), 231 (2025)*

CMB Distance Priors uses derived parameters (e.g., R , I_A , $\Omega_b h^2$) and Planck 18 uses derived parameters (e.g., ω_b , ω_{cb} , $D_M(z_*)/r_d$) *L. Chen, Q.-G. Huang, and K. Wang, JCAP 2019(02), 028., and E. Aubourg et al., Phys. Rev. D 92(12), 123516 (2015).*

Λ CDM Model Fails !!!



Beyond Λ : Alternative Models

How we explain this accelerated expansion?

Dark Energy

- Λ CDM
- *Quintessence*
- *Phantom*
- *Holographic*
- Chaplygin gas
-

Modified Gravity

- $f(R)$
- $f(T)$
- $f(Q)$
- DGP
- Horndeski
-

Thank you for your attention.

Any Questions?